
Education

- 2018 **Doctor of Philosophy**, *Georgia Institute of Technology*, Atlanta, GA, USA
Computational Science and Engineering
Thesis: *Scalable and Resilient Sparse Linear Solvers*
Advisor: Dr. Richard Vuduc
- 2016 **Master of Science**, *Georgia Institute of Technology*, Atlanta, GA, USA, GPA-4.0/4
Computational Science and Engineering
- 2011 **Master of Technology**, *Indian Institute of Technology, Madras*, India, GPA-8.5/10
Electrical Engineering, Specialization: Microelectronics and VLSI design
- 2011 **Bachelor of Technology**, *Indian Institute of Technology, Madras*, India, GPA-8.5/10
Electrical Engineering, Minor: Theoretical Computer Science

Experience

- August 2019–Present **Staff Scientist**, *Computational Data Analytics Group*, Oak Ridge National Laboratory, Oak Ridge, TN
- August 2018–July 2019 **Postdoctoral Research Associate**, *Computer Science Research Group*, Oak Ridge National Laboratory, Oak Ridge, TN
- 2011–2018 **Graduate Research Assistant**, *Georgia Institute of Technology*, Atlanta, GA
- May–August 2014 **Summer Intern**, *Parallel Computing Lab*, Intel Corporation, Santa Clara, CA
- May–August 2013 **Summer Intern**, *Computational Research Division*, Lawrence Berkeley National Laboratory, Berkeley, CA

Awards & Honors

Professional Awards

- 2022 **ORNL Special Performance Award**, For outstanding research contributions in the Computer Science and Mathematics Division
- 2022 **SC22 Gordon Bell Finalist**, ([Media](#)), Finalist for submission "Exaflops biomedical knowledge graph analytics"
- 2022 **SIAM PP22 Best Paper Prize**, ([Link](#)), Winner of the SIAM Activity Group on Supercomputing Best Paper Prize; ([SIAM News](#)), ([ORNL News](#))
- 2021 **R&D 100 Award Finalist**, ([Link](#))
- 2020 **SC20 Gordon Bell Finalist**, ([Link](#)), Finalist for submission "Scalable Knowledge Graph Analytics at 136 PetaFlop/s"
- 2019 **ORNL Outstanding Postdoctoral Research Associate**, For outstanding research contributions in the Computer Science and Mathematics Division
- 2019 **Graph500**, ([Link](#)), Member of the technical team that placed the Summit Supercomputer at ORNL 4th in the prestigious Graph500 List

Conference Travel Awards

- 2016 **HPDC**, Travel award recipient
- 2014 **SIAM Parallel Processing**, Travel award recipient
- 2014 **Copper-Mountain**, Travel award recipient

- 2015 **IPDPS**, Travel award recipient
Academic Awards
- 2006 **Indian National Mathematics Olympiad**, Selected as one of 35 awardees (100th percentile)
- 2006 **All India Rank 651 in IIT-JEE**, Top 0.2 percent of 300,000 applicants
- 2006 **National Standard Examination in Physics**, ([Link](#)), Awarded to top 1 percent (99th percentile)
Undergraduate Scholarship
- 2009 **DAAD-WISE**, Recipient of German Academic Exchange Service-Working Internships in Science and Engineering Scholarship (awarded to approximately 300 undergraduate students in India)
- 2010 **CRUISE fellowship**, Awardee of the CRUISE (Computing Research Undergraduate Intern Summer Experience) fellowship. Site: <http://www.cse.gatech.edu/research/cruise>

Publications

Google Scholar [Profile](#); Accessed on August 14, 2026

Metrics Citations: 1111 (Since 2021: 447); h-index: 12 (Since 2021: 11); i10-index: 17 (Since 2021: 14)

Conference Papers

- [1] Benjamin Cobb, Ramakrishnan Kannan, Konstantin Pieper, Piyush Sao, Yongseok Soh, Jee W Choi, Richard Vuduc, and Haesun Park. “Fast Active-Set Thresholding Method for Nonnegative Least Squares.” In: *2025 IEEE International Conference on Big Data (BigData)*. IEEE, 2025, pp. 856–867.
- [2] Piyush Sao, Andrey Prokopenko, and Damien Lebrun-Grandié. “PANDORA: A Parallel Dendrogram Construction Algorithm for Single Linkage Clustering on GPU.” In: *Proceedings of the 53rd International Conference on Parallel Processing*. 2024, pp. 908–918.
- [3] Yongseok Soh, Ramakrishnan Kannan, Piyush Sao, and Jee Choi. “Accelerated Constrained Sparse Tensor Factorization on Massively Parallel Architectures.” In: *Proceedings of the 53rd International Conference on Parallel Processing*. 2024, pp. 107–116.
- [4] Yang Liu, Nan Ding, Piyush Sao, Samuel Williams, and Xiaoye Sherry Li. “Unified Communication Optimization Strategies for Sparse Triangular Solver on CPU and GPU Clusters.” In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. 2023, pp. 1–15.
- [5] Hao Lu, Piyush Sao, Michael Matheson, Ramakrishnan Kannan, Feiyi Wang, and Thomas Potok. “Optimizing Communication in 2D Grid-Based MPI Applications at Exascale.” In: *Proceedings of the 30th European MPI Users’ Group Meeting*. 2023, pp. 1–11.
- [6] Piyush Sao and Xiaoye Sherry Li. “Brief Announcement: Communication Optimal Sparse LU Factorization for Planar Matrices.” In: *Proceedings of the 35th ACM Symposium on Parallelism in Algorithms and Architectures*. 2023, pp. 427–430.
- [7] Ramakrishnan Kannan, Piyush Sao, Hao Lu, Jakub Kurzak, Gundolf Schenk, Yongmei Shi, Seung-Hwan Lim, Sharat Israni, Vijay Thakkar, Guojing Cong, et al. “Exaflops biomedical knowledge graph analytics.” In: *2022 SC22: International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*. IEEE Computer Society. 2022, pp. 61–71.
- [8] Andrey Prokopenko, Piyush Sao, and Damien Lebrun-Grandié. “A single-tree algorithm to compute the Euclidean minimum spanning tree on GPUs.” In: *Proceedings of the 51st International Conference on Parallel Processing*. 2022, pp. 1–10.
- [9] Piyush Sao, Hao Lu, Ramakrishnan Kannan, Vijay Thakkar, Richard Vuduc, and Thomas Potok. “Scalable All-pairs Shortest Paths for Huge Graphs on Multi-GPU Clusters.” In: *Proceedings of the 30th International Symposium on High-Performance Parallel and Distributed Computing*. 2021, pp. 121–131.
- [10] Catherine D Schuman, Bill Kay, Prasanna Date, Ramakrishnan Kannan, Piyush Sao, and Thomas E Potok. “Sparse Binary Matrix-Vector Multiplication on Neuromorphic Computers.” In: *2021 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. IEEE. 2021, pp. 308–311.

- [11] Ramakrishnan Kannan, Piyush Sao, Hao Lu, Drahomira Herrmannova, Vijay Thakkar, Robert Patton, Richard Vuduc, and Thomas Potok. “Scalable knowledge graph analytics at 136 petaflop/s.” In: *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*. IEEE. 2020, pp. 1–13.
- [12] Piyush Sao, Ramakrishnan Kannan, Prasun Gera, and Richard Vuduc. “A supernodal all-pairs shortest path algorithm.” In: *Proceedings of the 25th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming*. 2020, pp. 250–261.
- [13] Piyush Sao, Christian Engelmann, Srinivas Eswar, Oded Green, and Richard Vuduc. “Self-stabilizing Connected Components.” In: *2019 IEEE/ACM 9th Workshop on Fault Tolerance for HPC at eXtreme Scale (FTXS)*. IEEE. 2019, pp. 50–59.
- [14] Piyush Sao and Ramakrishnan Kannan. “Multifrontal Non-negative Matrix Factorization.” In: *International Conference on Parallel Processing and Applied Mathematics*. Springer. 2019, pp. 543–554.
- [15] Piyush Sao, Ramakrishnan Kannan, Xiaoye Li, and Richard Vuduc. “A Communication-avoiding 3D Sparse Triangular Solve Algorithm.” In: *International Conference on Supercomputing*. Phoenix AZ, Accepted, June 2019.
- [16] Piyush Sao, Xiaoye S. Li, and Richard Vuduc. “A communication-avoiding 3D LU factorization algorithm for sparse matrices.” In: *Proceedings of the IEEE International Parallel and Distributed Processing Symposium (IPDPS)*. Vancouver, BC, Canada, May 2018.
- [17] Piyush Sao, Oded Green, Chirag Jain, and Richard Vuduc. “A Self-Correcting Connected Components Algorithm.” In: *Proceedings of the ACM Workshop on Fault-Tolerance for HPC at Extreme Scale*. ACM. 2016, pp. 9–16.
- [18] Piyush Sao, Xing Liu, Richard Vuduc, and Xiaoye Li. “A Sparse Direct Solver for Distributed Memory Xeon Phi-accelerated Systems.” In: *Parallel and Distributed Processing Symposium (IPDPS), 2015 IEEE International*. IEEE. 2015, pp. 71–81.
- [19] Piyush Sao, Richard Vuduc, and Xiaoye Sherry Li. “A distributed CPU-GPU sparse direct solver.” In: *European Conference on Parallel Processing*. Springer. 2014, pp. 487–498.
- [20] Piyush Sao and Richard Vuduc. “Self-stabilizing iterative solvers.” In: *Proceedings of the Workshop on Latest Advances in Scalable Algorithms for Large-Scale Systems*. ACM. 2013, p. 4.

Journal Articles and Preprints

- [21] Piyush Sao. “Fast Evaluation of Truncated Neumann Series by Low-Product Radix Kernels.” In: *arXiv preprint arXiv:2602.11843* (2026). URL: <https://arxiv.org/abs/2602.11843>.
- [22] Piyush Sao. “Ghosts of Softmax: Complex Singularities That Limit Safe Step Sizes in Cross-Entropy.” In: *arXiv preprint arXiv:2603.13552* (2026). URL: <https://arxiv.org/abs/2603.13552>.
- [23] Piyush Sao. “What Trace Powers Reveal About Log-Determinants: Closed-Form Estimators, Certificates, and Failure Modes.” In: *arXiv preprint arXiv:2601.12612* (2026). URL: <https://arxiv.org/abs/2601.12612>.
- [24] Piyush Sao, Narasinga Miniskar, Pedro Valero-Lara, Keita Teranishi, and Sudip Seal. “A Second-Moment Theory for Floating-Point Reduction Trees.” In: *arXiv preprint arXiv:2607.18758* (2026). URL: <https://arxiv.org/abs/2607.18758>.
- [25] Piyush Sao, Narasinga Miniskar, Pedro Valero-Lara, Keita Teranishi, and Sudip Seal. “Contraction-Gauge Preconditioning for Quantized Matrix Multiplication.” In: *arXiv preprint arXiv:2607.18745* (2026). URL: <https://arxiv.org/abs/2607.18745>.
- [26] Ahmad Abdelfattah, Willow Ahrens, Hartwig Anzt, Chris Armstrong, Ben Brock, Aydin Buluc, Federico Busato, Terry Cojean, Tim Davis, Jim Demmel, et al. “Interface for Sparse Linear Algebra Operations.” In: *arXiv preprint arXiv:2411.13259* (2024).
- [27] Xiaoye S Li, Paul Lin, Yang Liu, and Piyush Sao. “Newly Released Capabilities in Distributed-memory SuperLU Sparse Direct Solver.” In: *ACM Transactions on Mathematical Software* (2023).
- [28] Prasun Gera, Hyojong Kim, Piyush Sao, Hyesoon Kim, and David Bader. “Traversing large graphs on GPUs with unified memory.” In: *Proceedings of the VLDB Endowment* 13.7 (2020), pp. 1119–1133.
- [29] Piyush Sao, Xiaoye S Li, and Richard Vuduc. “A communication-avoiding 3D algorithm for sparse LU factorization on heterogeneous systems.” In: *Journal of Parallel and Distributed Computing* (2019).

- [30] Dongryeol Lee, Piyush Sao, Richard Vuduc, and Alexander G Gray. “A distributed kernel summation framework for general-dimension machine learning.” In: *Statistical Analysis and Data Mining: The ASA Data Science Journal* 7.1 (2014), pp. 1–13.

Books, Theses, Technical Reports, and Other Publications

- [31] Julian Bellavita, Piyush Sao, and Ramakrishnan Kannan. *Forward Error Bounds and Efficient Algorithms for Computing a Tensor Times Matrix Chain in Low Precision on GPUs*. SC25 poster. 2025. URL: https://sc25.supercomputing.org/proceedings/posters/poster_files/post166s2-file3.pdf.
- [32] Christian Engelmann, Rizwan Ashraf, Saurabh Hukerikar, Mohit Kumar, and Piyush Sao. *Resilience Design Patterns: A Structured Approach to Resilience at Extreme Scale (Version 2.0)*. Tech. rep. ORNL/TM-2022/2809. Oak Ridge National Laboratory, 2022. DOI: [10.2172/1922296](https://doi.org/10.2172/1922296).
- [33] Jeffrey A Graves, Thomas F Blum, Piyush Sao, Miaofang Chi, and Ramakrishnan Kannan. “FUNNL: Fast Nonlinear Nonnegative Unmixing for Alternate Energy Systems.” In: *Knowledge-Guided Machine Learning*. Chapman and Hall/CRC, 2022, pp. 261–286.
- [34] Vijay Thakkar, Ramakrishnan Kannan, Piyush Sao, Hao Lu, Drahomira Herrmannova, Robert Patton, Richard Vuduc, and Thomas Potok. *Dense Semiring Linear Algebra on Modern CUDA Hardware*. SIAM Conference on Computational Science and Engineering. 2021. URL: <https://graphblas.org/GraphBLAS-Pointers/Slides/cuASR.pdf>.
- [35] Piyush Sao. “Scalable and Resilient Sparse Linear Solvers.” PhD thesis. Georgia Institute of Technology, Aug. 2018. URL: <https://repository.gatech.edu/entities/publication/fc63f170-c9ef-4df4-a628-a34bbfa7186f>.
- [36] Piyush Sao. “Model Order Reduction Techniques for VLSI Circuit Simulation.” MA thesis. IIT Madras, May 2011.
- [37] Xiaoye S Li, James W Demmel, John R Gilbert, Laura Grigori, Piyush Sao, Meiyue Shao, and Ichitaro Yamazaki. *SuperLU Users’ Guide*. Tech. rep. Lawrence Berkeley National Laboratory, 1999.

Patents and Invention Disclosures

- 2025 **Knowledge graph analytics kernels in high performance computing**, Kannan, R., Sao, P. K., Lu, H., Herrmannova, D., Thakkar, V., Patton, R. M., Vuduc, R. W., & Potok, T. E., US Patent 12,417,246, Granted
Google Patents
<https://patents.google.com/patent/US12417246B2/en>
Application filed by UT-Battelle, LLC: July 30, 2021
- 2022 **U.S. Copyright Registration Certificate**, Title: *DSNAPSHOT*, Registration Number: TXu 2-314-193, Effective Date: 2022-05-05, ORNL
Authors:
 - Drahomira Herrmannova
 - Ramakrishnan Kannan
 - Hao Lu
 - Robert M Patton
 - Thomas E Potok
 - Piyush K Sao

Research Grants and Funding

- 2020–2023 **Principal Investigator**, Next generation sparse solvers *STRUMPACK/SUPERLU*, Exascale Computing Project, Department of Energy, \$480,000 USD
- 2022–2027 **Co-Principal Investigator**, *Sparsitute: Mathematical Multifaceted Integrated Capability Center (MMICC)*, Office of Advanced Scientific Computing Research, Department of Energy, \$2.5 million USD
<https://sparsitute.lbl.gov/people>
- 2020–2024 **Co-Principal Investigator**, *ASCENDS-II (Advances in Machine Learning to Improve Scientific Discovery at Exascale and Beyond)*, Department of Energy, \$1,200,000
Contributing to the development of advanced machine learning techniques to enable scientific discovery on exascale computing systems.

Talks and Posters

- SPAA'23 **Communication Optimal Sparse LU Factorization for Planar Graphs**, *Orlando, FL*, Jun. 16-19, 2023
- ICIAM'23 **Avoiding Communication in Sparse Graph All Pair Shortest Path Problem**, *Tokyo, Japan*
- SIAM CSE'23 **Communication-Avoiding Algorithms for Sparse Triangular Matrices**, *Feb. 26-Mar. 3, 2023*, Amsterdam, Netherlands
- SIAM PP'22 **A Communication-Avoiding 3D Algorithm for Sparse LU Factorization on Heterogeneous Systems**, *Invited Plenary Talk; Winner, SIAG Supercomputing Best Paper Award 2022*, Virtual (Seattle, WA)
- HPDC'21 **Scalable All-pairs Shortest Paths for Huge Graphs on Multi-GPU Clusters**, *Virtual*
- SIAM PP'20 **Communication-Avoiding Sparse Direct Solvers for Linear Systems & Graph Problems**, *Feb. 12-15, 2020*
- PPoPP'20 **A Supernodal All-pairs Shortest Path Algorithm**, *San Diego, CA*, Feb. 2020
- PPAM'19 **Multifrontal Non-negative Matrix Factorization**, *Bialystok, Poland*, Sep. 8-11, 2019
- FTXS'19 **Self-stabilizing Connected Components**, *Denver, CO*
- ICS'19 **A Communication-Avoiding 3D Sparse Triangular Solver**, *Phoenix, AZ*, Jun. 26-28
- SIAM CSE'17 **A Self-Correcting Connected Components Algorithm**, *Atlanta, GA*
- SIAM CSE'17 **Scalable Sparse Direct Solver for Hybrid Architectures**, *Atlanta, GA*
- CM'16 **Scalable Sparse Direct Solver for Hybrid Architectures**, *Copper Mountain, CO*
- FTXS'16 **A Self-Correcting Connected Components Algorithm**, *Kyoto, Japan*
- IPDPS'15 **A Sparse Direct Solver for Distributed Memory Xeon Phi-accelerated Systems**, *Hyderabad, India*
- SIAM LA'15 **A Sparse Direct Solver for Distributed Memory GPU and Xeon-Phi Accelerated Systems**, *Atlanta, GA*
- SC'13 **Self-stabilizing Iterative Solvers**, *Denver, CO*
- PMAA'14 **Self-stabilizing Iterative Solvers**, *Lugano, Switzerland*
- CM'14 **Self-stabilizing Iterative Solvers**, *Copper Mountain, CO*
- SIAM PP'14 **Self-stabilizing Iterative Solvers**, *Portland, OR*
- EuroPar'14 **A Distributed CPU-GPU Sparse Direct Solver**, *Porto, Portugal*
- IGWA'09 **Discretization of Convection-Diffusion Type Equation**, *IIT-Roorkee, India*

Professional Activities

Reviewer for

- International Journal of High Performance Computing Applications (IJHPCA)
- ACM Transactions on Parallel Computing
- SIAM Journal on Scientific Computing (SISC)
- Parallel Computing Journal (ParCO)
- ACM Transactions on Mathematical Software (ACM TOMS)
- Journal of Computational Science (JoCS)

Program Committee Member

- Supercomputing: SC20, SC22
- International Parallel and Distributed Processing Symposium: IPDPS 2021, IPDPS 2022, IPDPS 2023
- International Conference on Parallel Processing: ICPP 2022
- ACM International Conference on Supercomputing (ICS): ICS 2022, ICS 2023
- International Conference on Supercomputing: ISC 2022 - Posters
- Smoky Mountains Computational Sciences and Engineering Conference: SMC 2021, SMC 2022, SMC 2023
- High-Level Parallel Programming Models and Supportive Environments: HIPS 2022

Professional Society Memberships

- Society for Industrial and Applied Mathematics (SIAM)
- Association for Computing Machinery (ACM)
- Institute of Electrical and Electronics Engineers (IEEE)

Computer skills

Languages	C, C++, Python
Parallel	MPI, OpenMP, CUDA
Libraries	MATLAB, NumPy, pandas, LAPACK
Visualization	MATLAB, Matplotlib, D3.js, Graphviz, Inkscape
General purpose	SQL, LaTeX, Git